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1.Executive Summary

This report presents a comprehensive analysts of the delivery operation of a global e-commerce company managing and-to-end order fulfilment across multiple regions. The analysis covers 172.768 ceders January 2015 through January 2018, focusing on identifying tot causes of chronicle quantifying their financial impact, and establishing a data-driven frameworks for improvement

The headline finding la stark: 54.71% of all orders are delivered late, costing the business 12. 1M in eroded profitability on delayed orders alone. While total profit across full orders reached 57.5M, the persistent late-delivery problem represents both a significant financial drain and a major customer experience risk. A Random Forest predictive model achieved 74% accuracy in identifying at risk orders before they become late.

Key Finding	Metric
Overall Late Delivery Rate	54.71%
On-Time Delivery Rate	45.29%
Total Orders Analyzed	172,765
Total Profit (profitbale orders)	\$7.5M
Profit at Risk (delayed orders)	\$2.1M
Predictive Model Accuracy (RF)	74%

2.Business Context & Objectives

The company operates a global e-commerce platform selling products across categories including sporting good, fitness equipment, outdoor gear, footwear, and apparel across multiple international regions.

Core Business Problem

Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust reduced order profitability, and an inability to make lle commitments to buyers af point of putuse

Analytical Objectives

- Understand the current state of delivery performance across all dimensions (region, mode time segment)
- Quantify the financial impact of delays on order profitability
- Identify the primary operational bottlenecks driving late deliveries
- Build a predictive model to flag high-risk orders before they are shipped
- Deliver actionable recommendations to improve on the delivery and profitability

3.Key Performance Indicators (KPIs)

The following KPIs were computed directly from the cleaned dataset to establish a performance baseline.

172,765 Total Orders	94,523 Late Deliveries	54.71% Late Delivery %	45.29% On-Time Delivery%
\$7.5M	\$2.1M	3 days	\$22.03

Total Profit (profitable orders)	Profit at Risk (delayed orders)	90 th Percentile Delay	Avg Order Profit
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Late Delivery Rate at 54.71%	More than half of all orders arrive late. This is not an edge-case problem – it is the default experience for the majority of customer
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\$2.1M Profit at Risk	Orders that experienced delays collectively generated \$2.1M in profit that is under constant pressure from further operational deterioration
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90th Percentile Delay =3 Days	Even the most extreme cases are contained within 3 days of lateness, suggesting the problem is systemic and process-driven rather than catastrophic.
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4. Profitability Analysis

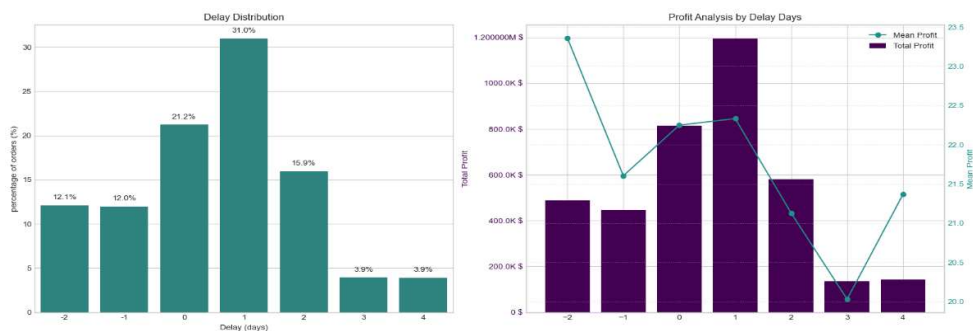
4.1 Profitability Distribution

Order-level profitability was classified into three tiers on Order profit Order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaning drag that is disproportionately concentrated among delayed shipment.



4.2 Delay Distribution & Profit vs. Delay Days

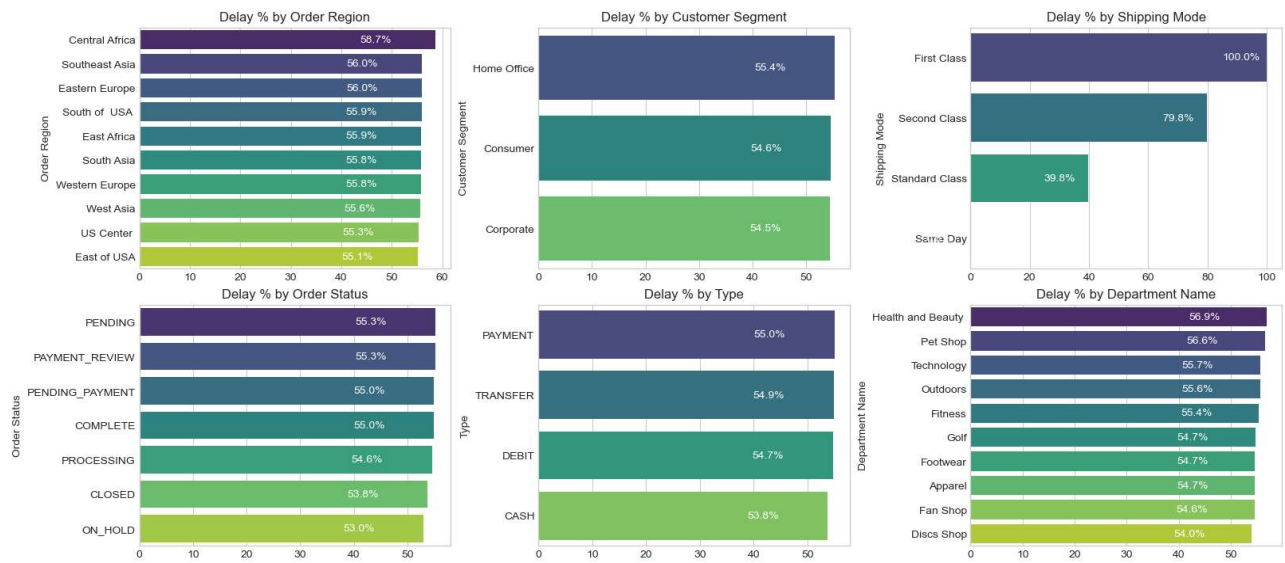
The delay distribution shows that 31.0% of all orders arrive 1 day late the single largest cohort. Combined, orders delayed by 1-4 days account for 54.7% of all order volume, directly mapping tom the overall late delivery rate.



A critical observation mean profit per order remains stable at approximately \$21-\$23 across all delay levels The profitability problem is therefore driven by volume of delayed orders, not by individual order economics making systemic throughput improvement the highest leverage intervention

5.Bottleneck Detection

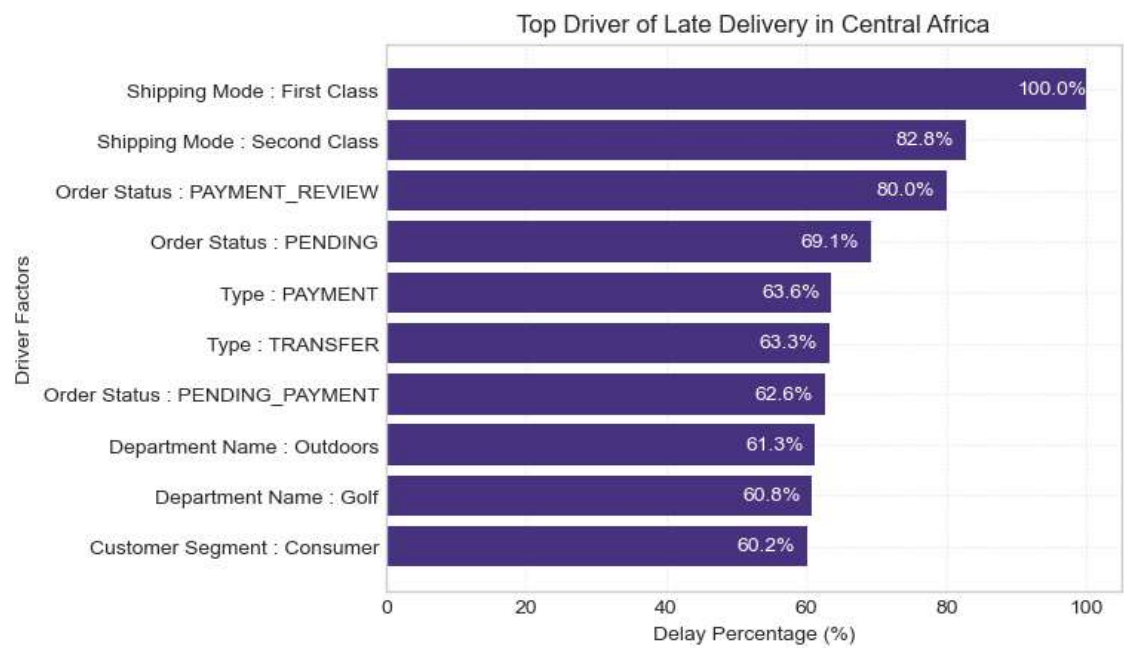
Delay percentage was computed across six key operational dimensions. The charts below reveal where the fulfilment process breaks down most severely.



Shipping Mode is the #1 Lever	First Class: !00% delay rate. Second Class:79.8%. Standard Class:39.8%. Same Day:0%. The mode assignment logic is the most impactful single variable to fix.
Regional Spread is Narrow (55-59%)	Central Africa leads at 58.7% but all regions cluster between 55-59%. This rules out a logicalized failure and confirms a company-wide systemic issues.
Customer Segments Nearly Identical (54.5-55.4%)	No segment receives preferential service. All customers – Consumer, Corporate, Home Office – experience the same broken delivery promise.
Health & Beauty (56.9%) and Pet Shop (56.6%) Lead by Department	These departments marginally outpace other in delay rate, warranting an inventory and carrier audit for these product categories.

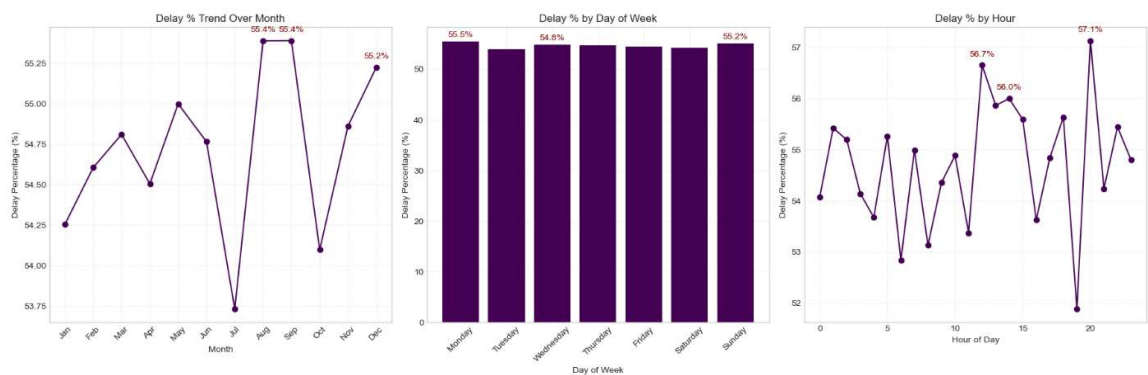
6. Root Cause Analysis

Central Africa was selected for deep-dive root cause analysis as the highest-delay region (58.7%). The chart below ranks the top 10 driver factors by delay percentage within this region.



7. Time-Based Delay Patterns

Three temporal dimensions were analysed month of year, day of week, and hour of day. While delay rates lively stable across a dimensions 153-571%), species peaks highlight opportunities for targeted capacity planning



Peak Delay Months: August & September (55.4%) and December (55.2%)	August and September are the highest months at 55.4%, with December close behind. These reflect mid-year promotions and Q4 holiday surge overwhelming fulfilment capacity. July represents a notable low-point (-53.75%), confirming seasonal variation is plannable.
Day-of-Week Variance is Minimal	Monday (55.5%) and Sunday (55.2%) are the peak days, but the spread across the week is less than 1 percentage point. Day-of-week is not a meaningful lever for operational intervention.
Intra-Day peaks: Hour 21 (57.1%), Hour 11 (56.7%), Hour 12 (56.0%)	Late-evening orders (hour 21) have the highest delay rate 57.1%, likely reflecting processing cutoff windows. Midday peaks suggest bottlenecks during peak-volume hours.

8. Machine Learning Model Results

A supervised classification model was built to predict whether an individual order will be delivered late (Late_delivery_risk = 1). The pipeline included frequency encoding of categorical variable, stratified train/test split, SMOTE oversampling to address class imbalance, and Random Forest Classification.

Class Imbalance Handling (SMOTE)

Split	Class 0 (On-time)	Class 1 (Late)
Before SMOTE (train)	59,030	79,182
After SMOTE (train)	79,182	79,182

Random Forest Classifier – Test Set Performance (34,553 records)

Metric	Class 0 (On time)	Class 1 (Late)	Overall (Weighted)
Precision	0.68	0.78	0.74
Recall	0.72	0.75	0.74
F1- Score	0.70	0.77	0.74
Accuracy	--	--	0.74
Support	14,758	19,795	34,553

74% Overall accuracy	The model correctly classifies nearly 3 in 4 orders – a strong foundation for production deployment as an order-level late-delivery alert system
Higher Precision on Late Orders (0.78)	When the model predicts an order will be late, it is correct 78% of the time – sufficient to trigger targeted intervention without generating excessive false alarms.
Recall for Late Order at 0.75	The model captured 75% of actual late orders. The 25% miss rate is acceptable for a first-generation alert system, with clear improvement potential
Class 0 Harder to Predict (Precision 0.68)	On-time deliveries share many characteristics with late ones given the systemic nature of delays, making them genuinely difficult to distinguish

9.Strategic Recommendations

Based on the finding across all analytical dimensions, the following prioritised action are recommended:

CRITICAL	Immediately Audit First Class & second Class Shipping Capacity First Class shipping has a 100% delay rate and Second Class 79.8%. These modes operate in complete contradiction to their brand promise. Conduct an immediate carrier SLA audit. Until resolve, consider suspending First Class or repricing to reflect
HIGH	Deploy the Predictive Alert System The Random Forest model (74% accuracy, 78% precision on late orders) is ready for production piloting. Integrate it into the order management system to flag high-risk order at confirmation and trigger proactive customer communication and priority handling queues.
MEDIUM	Develop Seasonal Surge Capacity Plans August, October (55.41%), and December (55.2%) are peak delay months. Build pre-season capacity plans including temporary logistics partnerships, inventory pre-positioning, and adjusted delivery promise windows
LOW	Continuously Retrain the Predictive Model Retrain the model quarterly. Explore adding carrier-level data, weather events, and warehouse utilisation features. Target a recall improvement to above 80% on late orders within 6 months

10.Conclusion

This analysis has surfaced a clear and urgent picture a global e-commerce operation where the majority of orders (54.71%) fail to meet their promised delivery windows, costing 52.1M in at-risk profit and undermining customer trust at scale.

The root causes are identifiable and addressable. Shipping mode misconfiguration (First Class at 100% late Second Class at 79.8%) is the dominant operational failure, Payment processing friction creates a secondary bottleneck. Seasonal volume spikes are not adequately planned for. And geographic disparities, while present, are secondary to the systemic company-wide pattern

A Random Forest predictive model trained on readily available order features already achieves 74% accuracy in identifying at-risk orders. This is a deployable tool, not a future aspiration.